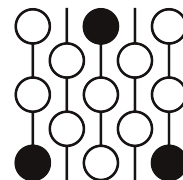


Review of Submission 9C

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

The proposed solution gives a combinatorial formula for the coefficients $c_\lambda(n)$ in terms of triples (r, π, t) , where r and t are integers and π is an ordered set partition.

The proof relies on Rhoades–Wilson’s formula [4] for the Hilbert series of the super coinvariant ring SR_n in terms of q -Stirling numbers. Using classical combinatorial results on Stirling numbers, together with elementary manipulations of q -identities, this Hilbert series is then rewritten as a generating function of decorated integer partitions.

Here are some comments:

- An important step in the argument is to use the universality of the coefficients $c_\lambda(n)$ which allows one to extract them from the Hilbert series of SR_n . This step does not seem to be explained clearly enough (see also the related comments on Lemma 3).
- Some references are missing, others are not sufficiently accurate.

Correctness. The combinatorial formula is correct, as are the main steps of the proof.

Novelty. The proof starts by relating the coefficients in question to the Hilbert series of the super coinvariant ring SR_n as expected, however it does not seem to cite the correct references.

The proof then uses standard references to connect this Hilbert series to the combinatorics of q -Stirling numbers. The trick used to factorize the series $H_n(q, z)$ (see Lemma 4) simplifies the final step of the proof.

The combinatorial interpretation in terms of ordered set partitions is then expected, however the use of the extra integer t makes the final formula feel slightly “less natural”.

2 Recommendation

The formula is correct and the solution is well written except for minor citation issues. I recommend publication after minor revisions.

3 Submission with inline comments

Abstract

Let $R_n^{(m)}$ be the diagonal coinvariant algebra of $\mathfrak{S}_n$ in m sets of n commuting variables, and write

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q_1, \dots, q_m).$$

For hook shapes $\lambda = (a+1, 1^b)$, we give a positive ordered-set-partition interpretation of $c_{\lambda}(n)$.

3.1 Statement of the hook formula

Reviewer 1

“Hook formula” is not a natural name here—it usually refers to a classical formula in combinatorial representation theory.

For $r, k \geq 1$, let $\text{OP}_{r,k}$ be the set of ordered set partitions

$$\pi = (B_1 \mid B_2 \mid \dots \mid B_k)$$

of $[r] = \{1, \dots, r\}$ into k nonempty blocks. Define

$$\text{inv}(\pi) = \#\{(i, j, x) : 1 \leq i < j \leq k, x \in B_i, x > \min(B_j)\}.$$

Also write $[k]_q = 1 + q + \dots + q^{k-1}$.

Theorem 1 (Hook coefficients). *Let $n \geq 1$. For $a, b \geq 0$, write the hook*

$$(a \mid b) := (a+1, 1^b).$$

Then

$$\sum_{a,b \geq 0} c_{(a \mid b)}(n) q^a z^b = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q \sum_{\pi \in \text{OP}_{r,k}} q^{\text{inv}(\pi)}.$$

Equivalently,

$$c_{(a+1, 1^b)}(n) = \#\left\{ (r, \pi, t) : \begin{array}{l} 1 \leq r \leq n-1, \pi \in \text{OP}_{r, r-b}, \\ 0 \leq t \leq r-b-1, \text{inv}(\pi) + t = a \end{array} \right\}.$$

In particular, $c_{(a+1, 1^b)}(n) = 0$ unless $0 \leq b \leq n-2$.

3.2 Ordered set partitions and q -Stirling numbers

Let

$$A_{r,k}(q) := \sum_{\pi \in \text{OP}_{r,k}} q^{\text{inv}(\pi)}.$$

Lemma 2. *The polynomials $A_{r,k}(q)$ satisfy*

$$A_{r,k}(q) = [k]_q \left(A_{r-1,k}(q) + A_{r-1,k-1}(q) \right),$$

with $A_{1,1}(q) = 1$. Hence

$$A_{r,k}(q) = [k]!_q \text{Stir}_q(r, k),$$

where $\text{Stir}_q(r, k)$ is the Carlitz q -Stirling number defined by

$$\text{Stir}_q(r, k) = [k]_q \text{Stir}_q(r-1, k) + \text{Stir}_q(r-1, k-1).$$

Proof. Remove the largest element r from an ordered set partition $\pi = (B_1 | \dots | B_k)$.

If r lies in a block B_i of size at least 2, removing r gives an ordered set partition in $\text{OP}_{r-1,k}$. Conversely, inserting r into block i of a partition in $\text{OP}_{r-1,k}$ creates exactly $k-i$ new inversions, since $r > \min(B_j)$ for every block $j > i$. Thus these insertions contribute

$$\sum_{i=1}^k q^{k-i} A_{r-1,k}(q) = [k]_q A_{r-1,k}(q).$$

If r is a singleton block, deleting that block gives a partition in $\text{OP}_{r-1,k-1}$. Conversely, inserting the singleton block $\{r\}$ in position i creates again $k-i$ new inversions. These insertions contribute $[k]_q A_{r-1,k-1}(q)$. Adding the two cases proves the recurrence. The identification with $[k]!_q \text{Stir}_q(r, k)$ follows from the same recurrence and initial condition. $\square$

3.3 Superspace and hook extraction

Let $\Omega_n = \mathbb{C}[x_1, \dots, x_n] \otimes \wedge(\theta_1, \dots, \theta_n)$, where the x_i commute and the θ_i anticommute. Let $\mathfrak{S}_n$ act diagonally by

$$w \cdot x_i = x_{w(i)}, \quad w \cdot \theta_i = \theta_{w(i)}.$$

The superspace coinvariant ring is

$$SR_n := \Omega_n / \langle (\Omega_n)_+^{\mathfrak{S}_n} \rangle.$$

Reviewer 1

I think that $(\Omega_n)_+$ has not been defined.

Let q track x -degree and z track θ -degree.

Lemma 3 (Hook specialization). *Let*

$$C_n(q, z) := \sum_{a, b \geq 0} c_{(a|b)}(n) q^a z^b.$$

Then

$$\text{Hilb}(SR_n; q, z) = 1 + (q + z)C_n(q, z).$$

Proof. The construction of diagonal coinvariants is functorial in the row-space.

This sentence is ambiguous.

Reviewer 1

For a purely even row-space of dimension m , its character is

$$\sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q_1, \dots, q_m).$$

Evaluating the same polynomial functor on a $1|1$ -dimensional superspace gives the superspace coinvariant ring SR_n .

A reference is missing here, I think that Lentfer 2025 should be cited for this result.

Reviewer 1

By the Berele–Regev hook Schur character theory, the ordinary Hilbert character of the Schur functor $\mathbb{S}_{\lambda}$ on one even variable q and one odd variable z is zero unless λ is a hook. Moreover,

$$h_{S_{(a|b)}}(q; z) = q^{a+1}z^b + q^a z^{b+1} = (q + z)q^a z^b,$$

and $h_{S_{\emptyset}}(q; z) = 1$. Therefore

$$\text{Hilb}(SR_n; q, z) = 1 + \sum_{a, b \geq 0} c_{(a|b)}(n)(q + z)q^a z^b = 1 + (q + z)C_n(q, z).$$

Give a reference for this formula.

Reviewer 1

This is the standard hook Schur specialization; see Berele–Regev [1] or Macdonald [3, Ch. I, §3, Ex. 23].

The references given are not the relevant ones here; this follows from the previous formula and the fact that the Hilbert series and the observation above on superspace coinvariant ring SR_n .

Reviewer 1

□

3.4 Hilbert series of SR_n

Rhoades and Wilson proved the following Hilbert series formula for the superspace coinvariant ring [4, Theorem 1.1 and Eq. (1.10)]:

Only Eq. (1.10) should be cited, Theorem 1.1 does not exist in [4].

$$\text{Hilb}(SR_n; q, z) = \sum_{k=1}^n z^{n-k} [k]_q! \text{Stir}_q(n, k).$$

Using the preceding lemma, this can be written as

$$\text{Hilb}(SR_n; q, z) = \sum_{k=1}^n z^{n-k} A_{n,k}(q).$$

Set

$$H_n(q, z) := \sum_{k=1}^n z^{n-k} A_{n,k}(q).$$

Then $H_1(q, z) = 1$.

Lemma 4. For $n \geq 2$,

$$H_n(q, z) - H_{n-1}(q, z) = (q + z) \sum_{k=1}^{n-1} z^{n-1-k} [k]_q A_{n-1,k}(q).$$

Consequently,

$$\frac{H_n(q, z) - 1}{q + z} = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q A_{r,k}(q).$$

Proof. Using the recurrence from the ordered-set-partition lemma,

$$A_{n,k} = [k]_q (A_{n-1,k} + A_{n-1,k-1}).$$

Therefore

$$\begin{aligned} H_n &= \sum_{k=1}^n z^{n-k} [k]_q A_{n-1,k} + \sum_{k=1}^n z^{n-k} [k]_q A_{n-1,k-1} \\ &= \sum_{k=1}^{n-1} z^{n-k} [k]_q A_{n-1,k} + \sum_{k=1}^{n-1} z^{n-1-k} [k+1]_q A_{n-1,k} \\ &= \sum_{k=1}^{n-1} z^{n-1-k} (z[k]_q + [k+1]_q) A_{n-1,k}. \end{aligned}$$

Since

$$z[k]_q + [k+1]_q = z[k]_q + 1 + q[k]_q = 1 + (q + z)[k]_q,$$

we obtain

$$H_n = H_{n-1} + (q + z) \sum_{k=1}^{n-1} z^{n-1-k} [k]_q A_{n-1,k}.$$

Telescoping from $H_1 = 1$ proves the second formula. $\square$

3.5 Proof of the theorem

By the hook-specialization lemma and the Rhoades–Wilson Hilbert series formula,

$$1 + (q + z)C_n(q, z) = H_n(q, z).$$

Thus

$$C_n(q, z) = \frac{H_n(q, z) - 1}{q + z}.$$

The preceding lemma gives

$$C_n(q, z) = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q A_{r,k}(q) = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q \sum_{\pi \in \text{OP}_{r,k}} q^{\text{inv}(\pi)}.$$

Since

$$[k]_q = \sum_{t=0}^{k-1} q^t,$$

the coefficient of $q^a z^b$ counts triples

$$(r, \pi, t)$$

such that $1 \leq r \leq n-1$, $\pi \in \text{OP}_{r,k}$, $k = r-b$, $0 \leq t \leq k-1$, and

$$\text{inv}(\pi) + t = a.$$

This is precisely the stated interpretation of $c_{(a+1, 1^b)}(n)$.

References

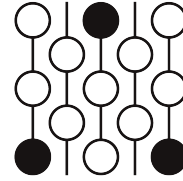
- [1] A. Berele and A. Regev, Hook Young diagrams with applications to combinatorics and to representations of Lie superalgebras, *Adv. Math.* **64** (1987), 118–175.
- [2] F. Bergeron, Multivariate diagonal coinvariant spaces for complex reflection groups, *Adv. Math.* **239** (2013), 97–108.
- [3] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, 1995.
- [4] B. Rhoades and A. T. Wilson, The Hilbert series of the superspace coinvariant ring, *Forum Math. Pi* **12** (2024), Paper No. e16.

Review of Submission 9C

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The mathematics in this paper is correct, though the proof of lemma 3 needs some work and added justification.

Novelty. This paper produces a combinatorial interpretation that, to my knowledge, is new. It predictably involves ordered set partitions, subject to a few self-contained conditions. That said, the conditions are read off straight from a q -series and the ordered set partitions are of different sizes, which does not make the interpretation very natural or satisfying. It is unlikely that this set of objects would appear elsewhere in the literature.

Lemma 4 is of particular note, as it cleverly factors $(q + z)$ out from the difference of consecutive Hilbert series, which allows for the construction of an equation directly relating the coefficients of the Hilbert series to the q -Stirling numbers, and thus ordered set partitions.

Attribution. This paper has several citation issues, including citing a theorem number that does not exist, citing papers that do not contain the result they should, and not acknowledging known results.

2 Recommendation

This paper is publishable with minor revisions. The citation issues and the proof of lemma 3 can easily be corrected.

3 Submission with inline comments

Abstract

Let $R_n^{(m)}$ be the diagonal coinvariant algebra of $\mathfrak{S}_n$ in m sets of n commuting variables, and write

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q_1, \dots, q_m).$$

For hook shapes $\lambda = (a + 1, 1^b)$, we give a positive ordered-set-partition interpretation of $c_{\lambda}(n)$.

3.1 Statement of the hook formula

For $r, k \geq 1$, let $\text{OP}_{r,k}$ be the set of ordered set partitions

$$\pi = (B_1 \mid B_2 \mid \cdots \mid B_k)$$

of $[r] = \{1, \dots, r\}$ into k nonempty blocks. Define

$$\text{inv}(\pi) = \#\{(i, j, x) : 1 \leq i < j \leq k, x \in B_i, x > \min(B_j)\}.$$

Reviewer 2

While technically correct, this is an odd way to define inversions on ordered set partitions.

Also write $[k]_q = 1 + q + \cdots + q^{k-1}$.

Theorem 1 (Hook coefficients). *Let $n \geq 1$. For $a, b \geq 0$, write the hook*

$$(a \mid b) := (a + 1, 1^b).$$

Then

$$\sum_{a,b \geq 0} c_{(a \mid b)}(n) q^a z^b = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q \sum_{\pi \in \text{OP}_{r,k}} q^{\text{inv}(\pi)}.$$

Equivalently,

$$c_{(a+1, 1^b)}(n) = \#\left\{ (r, \pi, t) : \begin{array}{l} 1 \leq r \leq n-1, \pi \in \text{OP}_{r, r-b}, \\ 0 \leq t \leq r-b-1, \text{inv}(\pi) + t = a \end{array} \right\}.$$

In particular, $c_{(a+1, 1^b)}(n) = 0$ unless $0 \leq b \leq n-2$.

3.2 Ordered set partitions and q -Stirling numbers

Let

$$A_{r,k}(q) := \sum_{\pi \in \text{OP}_{r,k}} q^{\text{inv}(\pi)}.$$

Lemma 2. *The polynomials $A_{r,k}(q)$ satisfy*

$$A_{r,k}(q) = [k]_q (A_{r-1,k}(q) + A_{r-1,k-1}(q)),$$

with $A_{1,1}(q) = 1$. Hence

$$A_{r,k}(q) = [k]_q! \text{Stir}_q(r, k),$$

where $\text{Stir}_q(r, k)$ is the Carlitz q -Stirling number defined by

$$\text{Stir}_q(r, k) = [k]_q \text{Stir}_q(r-1, k) + \text{Stir}_q(r-1, k-1).$$

Add an initial condition for the Carlitz q -Stirling numbers.

Proof. Remove the largest element r from an ordered set partition $\pi = (B_1 | \cdots | B_k)$.

If r lies in a block B_i of size at least 2, removing r gives an ordered set partition in $\text{OP}_{r-1,k}$. Conversely, inserting r into block i of a partition in $\text{OP}_{r-1,k}$ creates exactly $k - i$ new inversions, since $r > \min(B_j)$ for every block $j > i$. Thus these insertions contribute

$$\sum_{i=1}^k q^{k-i} A_{r-1,k}(q) = [k]_q A_{r-1,k}(q).$$

If r is a singleton block, deleting that block gives a partition in $\text{OP}_{r-1,k-1}$. Conversely, inserting the singleton block $\{r\}$ in position i creates again $k - i$ new inversions. These insertions contribute $[k]_q A_{r-1,k-1}(q)$. Adding the two cases proves the recurrence. The identification with $[k]_q! \text{Stir}_q(r, k)$ follows from the same recurrence and initial condition. $\square$

This is a known result from Steingrímsson and Remmel–Wilson and should be cited as such.

3.3 Superspace and hook extraction

Let $\Omega_n = \mathbb{C}[x_1, \dots, x_n] \otimes \wedge(\theta_1, \dots, \theta_n)$, where the x_i commute and the θ_i anticommute. Let $\mathfrak{S}_n$ act diagonally by

$$w \cdot x_i = x_{w(i)}, \quad w \cdot \theta_i = \theta_{w(i)}.$$

The superspace coinvariant ring is

$$SR_n := \Omega_n / \langle (\Omega_n)^{\mathfrak{S}_n}_+ \rangle.$$

Let q track x -degree and z track θ -degree.

Lemma 3 (Hook specialization). *Let*

$$C_n(q, z) := \sum_{a,b \geq 0} c_{(a|b)}(n) q^a z^b.$$

Then

$$\text{Hilb}(SR_n; q, z) = 1 + (q + z)C_n(q, z).$$

Proof. The construction of diagonal coinvariants is functorial in the row-space. For a purely even row-space of dimension m , its character is

$$\sum_{\lambda} c_{\lambda}(n) s_{\lambda}(q_1, \dots, q_m).$$

Which diagonal coinvariant, and which character is this referring to?

Evaluating the same polynomial functor on a $1|1$ -dimensional superspace gives the superspace coinvariant ring SR_n . By the Berele–Regev hook Schur character theory, the ordinary Hilbert character of the Schur functor $\mathbb{S}_\lambda$ on one even variable q and one odd variable z is zero unless λ is a hook.

Reviewer 2

This proof tries to sneakily go from $R_n^{(m)}$ to SR_n without proper justification. For the most direct fix, the author should just cite Lentfer’s “Diagonal supersymmetry for coinvariant rings.”

Moreover,

$$hs_{(a|b)}(q; z) = q^{a+1}z^b + q^a z^{b+1} = (q + z)q^a z^b,$$

and $hs_\emptyset(q; z) = 1$.

Reviewer 2

What is $hs_{(a|b)}$?

Therefore

$$\text{Hilb}(SR_n; q, z) = 1 + \sum_{a,b \geq 0} c_{(a|b)}(n)(q + z)q^a z^b = 1 + (q + z)C_n(q, z).$$

This is the standard hook Schur specialization; see Berele–Regev [1] or Macdonald [3, Ch. I, §3, Ex. 23]. $\square$

Reviewer 2

Neither of the two cited sources seems to contain this result.

3.4 Hilbert series of SR_n

Rhoades and Wilson proved the following Hilbert series formula for the superspace coinvariant ring [4, Theorem 1.1 and Eq. (1.10)]:

$$\text{Hilb}(SR_n; q, z) = \sum_{k=1}^n z^{n-k} [k]!_q \text{Stir}_q(n, k).$$

Reviewer 2

There is no Theorem 1.1 in this paper. There is an equation (1.10), correctly cited here, but it is a conjecture, not a result. The conjecture is proven later in the paper, as Theorem 5.3.

Using the preceding lemma, this can be written as

$$\text{Hilb}(SR_n; q, z) = \sum_{k=1}^n z^{n-k} A_{n,k}(q).$$

Set

$$H_n(q, z) := \sum_{k=1}^n z^{n-k} A_{n,k}(q).$$

Then $H_1(q, z) = 1$.

Lemma 4. For $n \geq 2$,

$$H_n(q, z) - H_{n-1}(q, z) = (q + z) \sum_{k=1}^{n-1} z^{n-1-k} [k]_q A_{n-1,k}(q).$$

Consequently,

$$\frac{H_n(q, z) - 1}{q + z} = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q A_{r,k}(q).$$

Proof. Using the recurrence from the ordered-set-partition lemma,

$$A_{n,k} = [k]_q (A_{n-1,k} + A_{n-1,k-1}).$$

Therefore

$$\begin{aligned} H_n &= \sum_{k=1}^n z^{n-k} [k]_q A_{n-1,k} + \sum_{k=1}^n z^{n-k} [k]_q A_{n-1,k-1} \\ &= \sum_{k=1}^{n-1} z^{n-k} [k]_q A_{n-1,k} + \sum_{k=1}^{n-1} z^{n-1-k} [k+1]_q A_{n-1,k} \\ &= \sum_{k=1}^{n-1} z^{n-1-k} (z[k]_q + [k+1]_q) A_{n-1,k}. \end{aligned}$$

Since

$$z[k]_q + [k+1]_q = z[k]_q + 1 + q[k]_q = 1 + (q + z)[k]_q,$$

we obtain

$$H_n = H_{n-1} + (q + z) \sum_{k=1}^{n-1} z^{n-1-k} [k]_q A_{n-1,k}.$$

Telescoping from $H_1 = 1$ proves the second formula. □

3.5 Proof of the theorem

By the hook-specialization lemma and the Rhoades–Wilson Hilbert series formula,

$$1 + (q + z)C_n(q, z) = H_n(q, z).$$

Thus

$$C_n(q, z) = \frac{H_n(q, z) - 1}{q + z}.$$

The preceding lemma gives

$$C_n(q, z) = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q A_{r,k}(q) = \sum_{r=1}^{n-1} \sum_{k=1}^r z^{r-k} [k]_q \sum_{\pi \in \text{OP}_{r,k}} q^{\text{inv}(\pi)}.$$

Since

$$[k]_q = \sum_{t=0}^{k-1} q^t,$$

the coefficient of $q^a z^b$ counts triples

$$(r, \pi, t)$$

such that $1 \leq r \leq n-1$, $\pi \in \text{OP}_{r,k}$, $k = r - b$, $0 \leq t \leq k-1$, and

$$\text{inv}(\pi) + t = a.$$

This is precisely the stated interpretation of $c_{(a+1,1^b)}(n)$.

References

- [1] A. Berele and A. Regev, Hook Young diagrams with applications to combinatorics and to representations of Lie superalgebras, *Adv. Math.* **64** (1987), 118–175.
- [2] F. Bergeron, Multivariate diagonal coinvariant spaces for complex reflection groups, *Adv. Math.* **239** (2013), 97–108.
- [3] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, 1995.
- [4] B. Rhoades and A. T. Wilson, The Hilbert series of the superspace coinvariant ring, *Forum Math. Pi* **12** (2024), Paper No. e16.